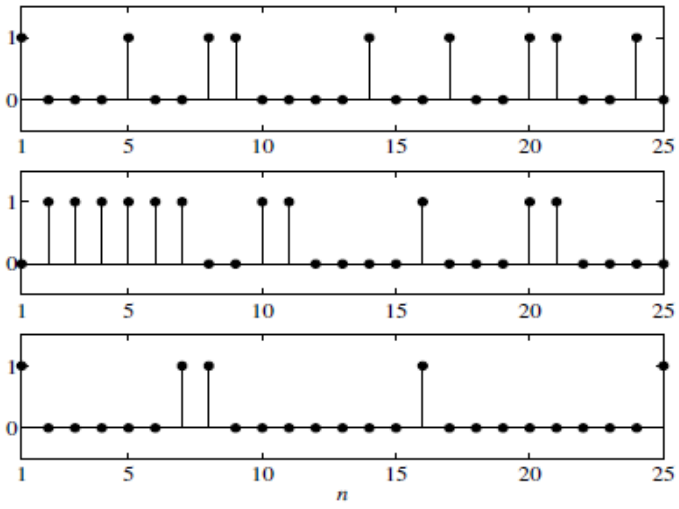
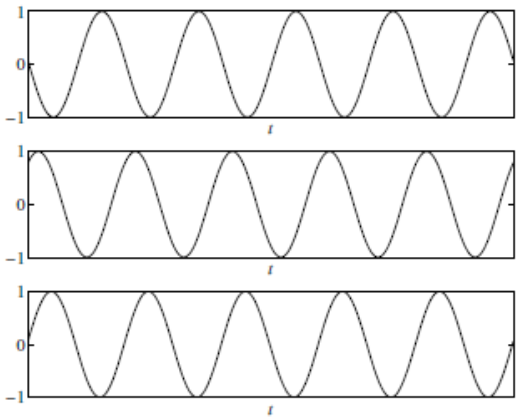
Tutorial 7

1. For each of the scenarios below, identify whether the random process is continuous-time or discrete-time. Then, identify the state space $\mathcal{S}$.

   (a) In sending a sequence of bits over a noisy channel, bits are flipped independently with probability p. Let $X_n = 1$ if the nth bit is flipped and $X_n = 0$ otherwise. Then $\{X_n, n = 1, 2, \dots\}$ is an i.i.d. Bernoulli(p) sequence. The figure below shows 3 possible realizations of this random process.



   (b) In radio communications, the carrier signal is often modeled as a sinusoid with a random phase. The reason for using a random phase is that the receiver does not know the time when the transmitter was turned on or the distance from the transmitter to the receiver. The mathematical model for this is a random process given by $X(t) = \cos(2\pi f t + \Theta)$, where f is the carrier frequency and $\Theta \sim U(-\pi, \pi)$. The figure below shows 3 possible realizations of this random process.



2. Let $\{N(t), t \in [0, \infty)\}$ be a Poisson process with rate $\lambda = 0.5$.

   (a) Find the probability of no arrivals in $(3, 5]$.

   (b) Let Y_1, Y_2, Y_3 and Y_4 be the number of arrivals in the intervals $(0, 1], (1, 2], (2, 3], (3, 4]$ respectively. Find the probability that there is exactly one arrival in each of the intervals.

   (c) Find the probability that there are two arrivals in $(0, 2]$ and three arrivals in $(4, 7]$.

   (d) Find the probability that there are two arrivals in $(0, 2]$ or three arrivals in $(4, 7]$.

   (e) What is the average time between arrivals?

   (f) Find the variance of the time it takes for the first n arrivals.

3. Two teams A and B play a soccer match. The number of goals scored by Team A is modeled by a Poisson process $N_1(t)$ with rate $\lambda_1 = 0.02$ goals per minute, and the number of goals scored by Team B is modeled by a Poisson process $N_2(t)$ with rate $\lambda_2 = 0.03$ goals per minute. The two processes are assumed to be independent. Let $N(t)$ be the number of goals in the game up to and including time t. The game lasts for 90 minutes.

   Except for Part (d), compute numerical answers and leave them in 2 decimal places for the following questions.

   (a) Find the probability that no goals are scored, that is, the game ends with a $0 - 0$ draw.

   (b) Find the probability that at least two goals are scored in the game.

   (c) Find the probability of the final score being Team A: 1 and Team B: 2.

   (d) Find the probability that they draw.

4. In a town, a drunken man is wandering along a street with many intersections. He does not really know which way he is going. He is currently at an intersection somewhere in the middle of the street. Suppose at the end of each block, he either goes north with probability p, or he goes south with probability $1-p$. A mathematical model for the movements of the drunken man is known as a **one-dimensional random walk**. Consider a stochastic process $\{X_n : n = 0, 1, \dots\}$ with state space $\{a, a+1, \dots, b\}$, a subset of integers, finite or infinite, and $a < b$. Suppose that if, for some n, $X_n = i$, then $X_{n+1} = i+1$ with probability p, and $X_{n+1} = i-1$ with probability $1-p$. Find the transition matrix of this Markov chain.

5. A computer is shared by 2 users who send tasks to a computer remotely and work independently. At any minute, any connected user may disconnect with probability 0.5, and any disconnected user may connect with a new task with probability 0.2. Let X_n be the number of concurrent users at time n (minutes). We now have a Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with $X_n \in \mathcal{S} = \{0, 1, 2\}$.

   (a) Draw the transition diagram and hence find the transition matrix for this Markov chain.

   (b) If both users are connected at 10:00, what is the probability that there will be no users at 10:02?

   (c) If we know that the 2 users will connect to the computer at 10:00, what is $\pi^{(0)}$? What is $\pi^{(2)}$?

   (d) Now, suppose that all states are equally likely at 10:00. Compute the probability of 1 connected user at 10:02.